

Zeyu Chang

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EDUCATION

University of Electronic Science and Technology of China

M.Eng. in Information and Communication Engineering

Sept. 2024 – Jun. 2027

Chengdu, China

Advisor: Prof. Xiaoling Zhang · **Co-advisor:** Assoc. Prof. Tianjiao Zeng

University of Electronic Science and Technology of China (UESTC)

B.Eng. in Communication Engineering · GPA: 3.56/4.00

Sept. 2020 – Jun. 2024

Chengdu, China

University of Glasgow

Dual-degree program at Glasgow College, UESTC

B.Eng. (Hons) in Electronics and Electrical Engineering · Upper Second-Class Honours (2:1)

RESEARCH INTERESTS

Computational Imaging Lensless Imaging Inverse Problems 3D Reconstruction

PUBLICATIONS & MANUSCRIPTS

Zeyu Chang*, Tianjiao Zeng, Xiaoling Zhang. “Unsupervised Lensless Image Reconstruction Driven by Structure Consistency Modeling in Low-Light Scenarios.” *Fifth International Computational Imaging Conference (CITA 2025)*, Proc. SPIE 14000, 1400022, 2026.

Yunqi Wang*, **Zeyu Chang**, Furui Liu, Edmund Lam, Tianjiao Zeng. “Robust Lensless Reconstruction under Unseen Imaging Conditions via Causal Deconfounding.” *Engineering Applications of Artificial Intelligence*. **Under Review**.

Yunqi Wang, Zhenxiao Zhou, Xu Zhan, **Zeyu Chang**, Yue Shi, Tianjiao Zeng*. “SenseGS: Compressive Lensless Novel View Synthesis via Sensor-Domain Self-Supervision.” *Optics and Laser Technology*. **Under Review**.

RESEARCH EXPERIENCE

Continuous-Depth PSF Modeling

Aug. 2026 – Present

University of Electronic Science and Technology of China

- Investigated the continuous axial evolution of depth-dependent PSFs to address the limitation of discrete-depth calibration for continuous-scene modeling.
- Analyzed measured PSFs in spatial and Fourier domains, revealing depth-dependent spatial scaling, corresponding magnitude-spectrum scaling, and phase signatures of spatial shifts.
- Developed a continuous-depth PSF approximation model from these relationships; preliminary validation shows close agreement with measured PSFs, with reconstruction evaluation ongoing.

Single-Shot Lensless 3D Reconstruction

Sept. 2025 – Present

University of Electronic Science and Technology of China

- Developed a single-shot lensless 3D method to jointly recover an all-in-focus image and depth map from one measurement without supervised training data.
- Represented the scene by intensity and depth-allocation variables that form depth-wise slices, then propagated them through calibrated depth-dependent PSFs to impose multi-depth forward-model consistency.
- Addressed depth ambiguity from highly correlated neighboring PSFs with image-structure-based semantic regularization and entropy regularization; outperformed the evaluated supervised baselines in simulated and experimental comparisons.

Low-Light Lensless Image Reconstruction

Mar. 2025 – Sept. 2025

University of Electronic Science and Technology of China

- Investigated low-light lensless reconstruction without supervised training data, targeting noise-sensitive per-measurement optimization under low-SNR measurements.
- Designed local-global structural consistency and adaptive partial convolution to preserve image structure while reducing the influence of noise-dominated regions.
- Introduced geometric equivariance for transformation-consistent reconstruction; outperformed the evaluated baselines on simulated and experimental low-light data, resulting in a first-author publication.

AWARDS & HONORS

- First-Class Academic Scholarship**, University of Electronic Science and Technology of China, 2025
- Outstanding Student Scholarship**, University of Electronic Science and Technology of China, 2022
- Silver Award**, Sichuan Provincial Challenge Cup College Student Entrepreneurship Competition, 2022

SKILLS

Programming & Frameworks: Python, MATLAB, PyTorch, LaTeX

Imaging & Reconstruction: PSF calibration and modeling, Fourier optics, optimization- and neural-network-based reconstruction

Simulation & Experiments: Blender-based scene simulation, optical system calibration, industrial-camera acquisition, translation stage control

Languages: Mandarin Chinese (native), English (IELTS 7.5)